

Measurement of optical fiber bending stiffness

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ABSTRACT

This paper presents traceable measurements of bending stiffness for standard optical SMF-28 fiber. Stiffness values were derived from force-displacement measurements, performed in a setup specifically designed for the stiffness calibration of MEMS and AFM cantilevers. As a test sample, fibers with free lengths of 2.8 mm, 5.0 mm, and 10.0 mm were examined. The correspondingly measured stiffness of 100.0 N/m, 14.6 N/m, and 2.4 N/m are broadly consistent with the Euler-Bernoulli beam theory. During the measurement process, deviations between load and unload cycles reaching 2 % were also observed, which can be caused by the method of fixing the optical fiber, its internal properties, stick-slip effects that occur between the sample and the load button, and internal properties of the system. This study highlights that the presented measurement approach can be used to improve the understanding of the mechanical properties of widely used optical fiber types and qualify them for use as tactile probes or for other mechanical sensing principles.

1. Introduction

Fiber optic sensors with their characteristic advantages such as their compact size, insensitivity to electromagnetic interference, and high sensitivity are constantly gaining importance in many fields ranging from biomedical to civil engineering. A vast field of different sensing principles based upon fiber optics allows various applications from molecule sensing [1] over pressure-sensing [2] to structural health monitoring for buildings [3]. Particularly noteworthy within the context of this paper is the concept of fiber optic shape sensing (FOSS), where in fiber strain measurements are used to reconstruct the shape of objects or the fiber itself. FOSS allows tool shape reconstruction for surgery robotics and endoscopic techniques and has proven itself for the monitoring of engineering structures [4], but is also promising for applications in precision metrology.

While there exist multiple approaches to the realization of FOSS e.g., strain sensing based on backscattering or FBG-based strain sensing [5], the concept of fiber segment interferometry [6] allows for highest strain resolutions and is therefore most suitable for high precision measurements. One possible application for high precision FOSS are tactile micro-probes for coordinate measuring machines. Previously glass fibers have already been commonly used as shaft material for those probes [7, 8]. In such systems the tip deflection needs to be measured. Until now, for most realizations of tactile probes with optical fiber shafts, tip deflection was measured using complex external sensors like photo-sensitive devices, interferometric or focus sensors [9].

However, in principle FOSS should allow for fiber tip deflection measurement within the fiber itself, yielding great potential for miniaturization of sensor heads. Tip displacement is not only measured to achieve accurate information about the position of the probe head in respect to the stages of the coordinate measuring machine, but also is needed to ensure a constant probing force. This needs to be tightly

controlled so that deformation of the sample is minimized, but effects of fluid films present on the surface are mitigated. For this reason, knowledge of the mechanical properties of the fiber, in particular the fiber bending stiffness, is of great importance, in order to obtain an accurate correlation between tip displacement measurement and probing force.

In this contribution, we present a series of traceable bending stiffness measurements of an SMF-28 optical fiber using a modified force-displacement measurement device developed at the Technische Universität Ilmenau to measure the stiffness of MEMS and cantilevers over a wide range [10,11]. Measuring the force of the sample on the device is possible due to electromagnetic or electrostatic force compensation principles (EMFC and ESFC). The range of measured force varies from 10 nN to 2 mN, for forces in the nN range and stiffness in the mN/m range, the use of an electrostatic force compensation principle is assumed. This article presents the results of testing a device for measuring optical fiber bending stiffness. Since the analytically calculated expected fiber stiffness are in the range of 1 N/m, the electromagnetic compensation principle was used for calibration. Furthermore, we present the basic theoretical calculations of fiber bending stiffness considering different coating types and shaft length. These explanations are content of the following section, followed by a description of the stiffness measurement setup.

2. Theoretical calculation of fiber bending stiffness

In the following section, we present a theoretical basis for the calculation of the bending stiffness of optical fiber. The construction of standard optical single mode fiber is depicted in Fig. 1. It consists of core (diameter of 9 μm), cladding (diameter of 125 μm) of silica and protective coating (diameter of $\approx 230 \mu\text{m}$). As coating materials usually acrylate or polyimide are being used. Core and cladding are both out of

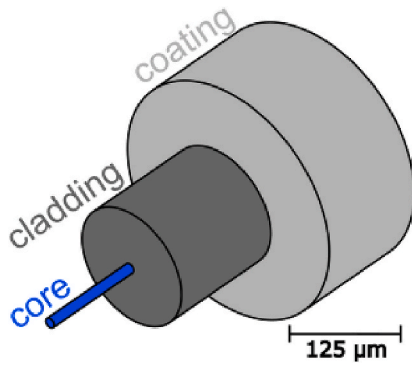


Fig. 1. Schematic of standard optical single mode fiber.

silica, however the core is doped with a material such as Germanium to reach variation in refractive index, which allows light propagation within the core.

Within this study the fiber is modelled using the Euler Bernoulli beam theory which is applicable for the small deflections used in the later experiments. Thereby fiber is represented by a cantilever configuration with one external force applied at the fiber tip and the other end placed in an ideal fixation. For that the tip deflection of the cantilever can be derived:

$$d = \frac{F l^3}{3 E I_z} \quad (1)$$

where F is the force applied to the fiber tip, l is the free shaft length of the fiber, E is the Young modulus of fiber and I_z is the area moment of inertia. The possible variations of mechanical properties between fiber core and cladding are neglected during this calculation, since core material is situated closest to the neutral axis of the fiber and therefore any slight variations in Young modulus due to the dopant will have virtually no impact on the mechanical behavior of the fiber. The coating material however affects the stiffness of the fiber significantly. When taking into account the coating and assuming ideal bonding between silica fiber and coating material, the stiffness of the fiber can be calculated to be:

$$k_{combined} = \frac{3}{l^3} (E_{silica} I_{z_{core}} + E_{cladding} I_{z_{cladding}}) \quad (2)$$

Results for the calculation of stiffness of stripped fiber (coating removed), fiber with acrylate coating and fiber with polyimide coating for different fiber shaft lengths are shown in Fig. 2. For these calculations, values of 68.56 GPa, 1,3 GPa and 3,6 GPa were used for the Youngs modulus of Silica, acrylate and polyimide respectively. Absolute

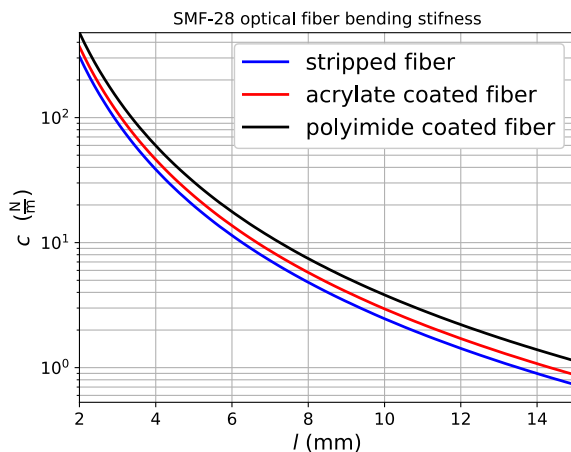


Fig. 2. Dependence of fiber bending stiffness in a cantilever configuration on its free shaft length (logarithmic scale along the y-axis).

Values for these calculations for fiber shaft lengths of 1.0 mm, 2.5 mm, 5.0 mm, 10.0 mm and 20.0 mm are also collected in Table 1. It can be seen that fiber stiffness increases drastically with reduced fiber shaft length l . Therefore, uncertainties of fiber shaft length will have bigger impact on the accuracy of the stiffness calculation for smaller shaft lengths.

3. Experimental setup and measurements

The force-displacement measurement device [11] is located in a vacuum chamber on an aluminium fixed platform. The main part of the device is an electromechanical balance with a hinge that is 50 μ m thick. To position the optical fiber relative to a load button with a flat tip with a diameter of 10 μ m, three linear platforms are used along the x-, y- and z-axes and a piezo drive along the z-axis (Physik Instrumente (PI) GmbH & Co. KG). The movement of the sample is recorded by a differential interferometer (SIOS Meßtechnik GmbH) with a resolution of 5 μ m. The operational concept of the device is illustrated in Fig. 3.

When a static force is applied to a monolithic aluminum alloy balance (2) through a slender connection like a hinge (5), it results in a displacement of the balance from its zero position. To measure this displacement, two position sensors are employed: a photo detector with a slit aperture (1) and the differential balance interferometer (10) (SIOS Meßtechnik GmbH), tailored to the setup with a resolution of 20 μ m. Following this, the displacement signal is fed into a microcontroller equipped with additional ADC/DAC peripherals. The microcontroller's output signal is then directed to one of two actuators: either a customized coil (7) attached to the balance within a constant magnetic field or an electrostatic differential capacitor consisting of two custom electrodes (4) and the balance itself. These actuators generate either electromagnetic or electrostatic forces, effectively counterbalancing the effect of the static force and restoring the balance to its zero position.

Since theoretical calculations of the measured stiffness of optical fibers (shaft lengths 2.8 mm, 5.4 mm and 10.0 mm) are above 1 N/m, the measurement principle based on electromagnetic force compensation was used in the measurement process. Examples of the use of the electrostatic principle of force compensation are given in Ref. [11]. Accordingly, in mathematical expression, the static force of the optical fiber on the balance is presented as follows:

$$F_{fiber} = B \cdot l \cdot i_{coil} \cdot \frac{r_{coil}}{r_{fiber}}, \quad (3)$$

where $B \cdot l$ is the electromagnetic force factor, depending on the magnetic-flux density and the length of the coil, i_{coil} is the compensated coil current measured through a shunt resistor with a multimeter, r_{coil} is the distance from the hinge to the coil, and r_{fiber} is the distance from the hinge to the load button. The electromagnetic force constant $B \cdot l$ was statically calibrated before measurements using a 10 mg wire reference weight.

For fixation within the force-displacement measurement device, SMF-28 optical fiber pieces were attached to custom 3-D printed adapters using Permabond 105 glue. The final samples had free shaft lengths of 2.8 mm, 5.4 mm and 10.0 mm. Shaft length thereby was only estimated by the length to diameter ratio in the microscope used during the glueing process and therefore is not accurate to less than 0.1 mm. An

Table 1

Calculated stiffness values for SMF-28 fiber.

Fiber shaft length l in mm	Stiffness k in N/m		
	Stripped fiber	Acrylate coated fiber	Polyimide coated fiber
1.0	2464.92	2953.91	3819.05
2.5	157.75	189.05	244.42
5.0	19.72	23.63	30.55
10.0	2.46	2.95	3.82
20.0	0.31	0.37	0.48

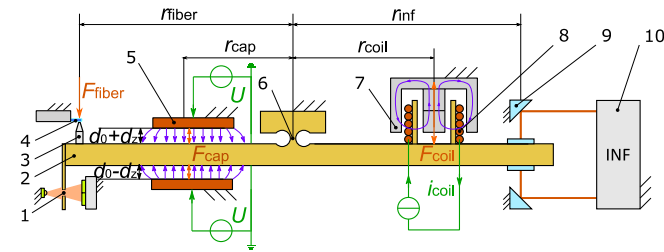


Fig. 3. The traceable force-displacement calibration principle: (1) slit aperture; (2) beam balance; (3) load button; (4) fiber; (5) electrode of a capacitor; (6) hinge; (7) permanent magnet; (8) coil; (9) deflection mirror; (10) interferometer [11].

example of the used samples is depicted in Fig. 4. Because standard SMF-28 fiber is only available with acrylate coating and bare fiber may be weakened by the stripping process only acrylate coated fiber was used for the experiments presented in this paper.

Visual assessment of the position of the optical fiber sample relative to the load button is carried out by using a camera with 12× zoom lens (Thorlabs). An example of the resulting image is shown in Fig. 5.

The stiffness measurements of optical fiber samples of different lengths took place in a vacuum chamber filled with air at normal atmospheric pressure. The z-axis path of the samples does not exceed 4 μm for all measurements so that real measurements can be compared with theoretical calculations based on Bernoulli's beam theorem. The step-wise effect of optical fiber on the balance has a pyramidal shape, in each cycle there are 5 steps of loading and unloading. The stiffness is calculated based on the sample interferometer data and the compensating current when using the fitting function, accordingly, the values of the first two and two last steps are not taken into account due to possible attractive and repulsive forces. Also, initially the fiber is close to the load button, but does not touch it. Fig. 6 shows the results of measuring the stiffness of an optical fiber with a shaft length of 2.8 mm.

The measurements lasted for 30 cycles and about 4 hours, the time of each loading and unloading step was 50 s. The measured stiffness is equal to 99.96 N/m with a standard deviation of 0.15 N/m. Fig. 6 (c) shows a clear difference in stiffness values between the loading and unloading cycles. The difference between the average load and unload values does not exceed 2 %. This difference can be caused by both sample hysteresis and system hysteresis. In addition, a PID controller was used for the control loop without additional adaptation for a different range of force compensation. Also, when measuring an optical fiber with a length of 2.8 mm, a pronounced stick-slip effect is observed.

Fig. 7 shows the results of measuring the stiffness of an optical fiber with a length of shaft 5.4 mm. Like the 10.0 mm fiber measurements, these measurements lasted 30 cycles, about 4 hours, and each step lasted 50 s. The measured stiffness is 14.56 N/m with standard deviation 0.02 N/m. The difference in load and unload values does not exceed 1.5 %. The stick-slip effect is significantly manifested. Also, the fiber optic sample twists when it applies to the load button. This behavior of the optical fiber during measurements may be caused by non-ideal fixation of the sample when using glue.

Fig. 8 shows the results of measuring fiber stiffness of a sample with a free shaft length of 10.0 mm. Each step of the cycle lasted 40 s, a total of 30 load and unload cycles were performed with a total time of about 3.5 hours. The measured stiffness is 2.43 N/m with a standard deviation of 0.02 N/m. Fig. 8 (c) shows that the difference between load and unload values is not pronounced compared to previous measurements of 2.8 mm and 5.4 mm fiber shaft lengths. The difference between the load and unload curves does not exceed 1.5 %.

Since the purpose of this article is to improve the understanding of the mechanical properties of optical fiber for its subsequent qualification when used as a tactile sensor, the calculation of measurement uncertainty according to the GUM is not performed at this stage and standard

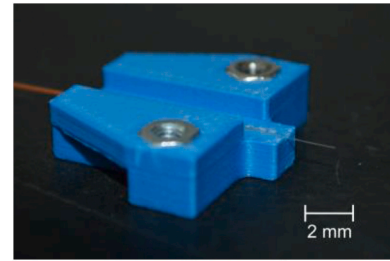


Fig. 4. SMF-28 optical fiber attached to custom holder for bending stiffness measurements.

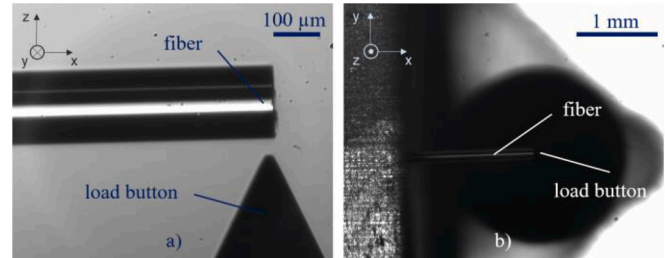


Fig. 5. Visual inspection of the 10 mm optical fiber under calibration in force-displacement measurement device: a) xz-plane, b) xy-plane.

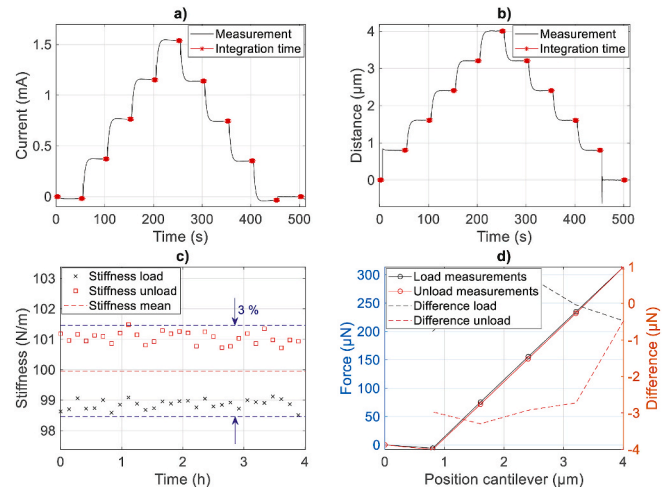


Fig. 6. Fiber SMF-28 with length 2.8 mm (EMFC measurements): a) measured signals of the compensation current with the marked-up integration time of one measurement cycle, b) corresponding measured fiber displacement with the marked-up integration time for one cycle, c) fiber stiffness measurement, d) force-displacement characteristic for one cycle.

deviation values are used to assess the accuracy of measurements.

4. Results and discussion

The summarized results of measurements and theoretical calculations are presented in Table 2. Deviations between theoretical calculations and fiber stiffness measurements reach 11 % for 2.8 mm, 7 % for 5.4 mm and 1.6 % for 10.0 mm shaft lengths.

However, when considering an estimated uncertainty of the shaft length determination of 0.1 mm, good agreement between measured values and theoretical calculations can be concluded.

Further sources of deviations between theoretical values and results could be the weakening of the optical fiber due to mechanical stress during the stripping process (removal of coating layer) and

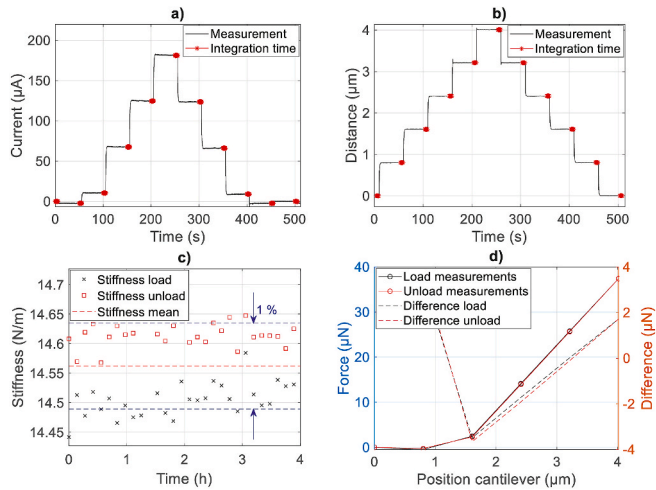


Fig. 7. Fiber SMF-28 with length 5.4 mm (EMFC measurements): a) measured signals of the compensation current with the marked-up integration time of one measurement cycle, b) corresponding measured fiber displacement with the marked-up integration time for one cycle, c) fiber stiffness measurement, d) force-displacement characteristic for one cycle.

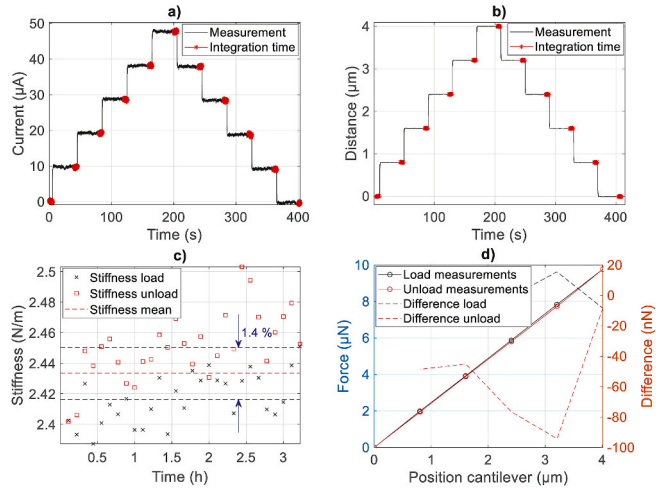


Fig. 8. Fiber SMF-28 with length 10.0 mm (EMFC measurements): a) measured signals of the compensation current with the marked-up integration time of one measurement cycle, b) corresponding measured fiber displacement with the marked-up integration time for one cycle, c) fiber stiffness measurement, d) force-displacement characteristic for one cycle.

Table 2

Comparative results of measurements and calculations of SMF-28 fiber stiffness.

Fiber shaft length l in mm	Measured stiffness in N/m	Theoretical calculation in N/m	Relative deviation in %
2.8	99.96	112.29	11
5.4	14.56	15.65	7
10.0	2.43	2.47	1.6

imperfections in the sample-holder configuration, such as elasticity of the glue and slight axial deviations between optical fiber and measurement device.

In conclusion, despite the double-digit percentage discrepancy between one of the measurements and the calculated data, these results are in reasonable agreement with the theory and also confirm the possibility of studying the mechanical properties of the fiber using a force-displacement measurement device. In this way the presented results

pave the way for future investigations using the established method for EMFC-based stiffness calibrations in fiber-optic research.

5. Outlook

Future experiments will include the use of multicore fibers (MCF) and simultaneous measurement of fiber tip deflection with fiber optic shape sensing (FOSS) based on the novel fiber segment interferometry (FSI) technique proposed in Ref. [12]. The comparison of measured values with the data from the force-displacement measurement device then is the basis for the first characterization of FSI in multicore fibers. Thereby the achievable tip displacement resolution as well as the force resolution should be quantified. Additionally, we hope to gain further insights into the interactions occurring between sample and balance from the self-sensing MCF samples. This research will be the basis for the realization of self-sensing multicore fiber probes for tactile sensing in micro coordinate measuring and various other sensing concepts building on the combination of FSI and MCF.

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